Problem 1

$$\int_0^1 x^{44} \ln(x) \, \mathrm{d}x$$

Problem 2

$$\int_0^{\frac{\pi}{4}} (1 - x \tan(x)) \tan(x) \, \mathrm{d}x$$

Problem 3

$$\int_{-1}^1 f_n(x) f_m(x) \, \mathrm{d}x \qquad \text{for positive integers } m, n,$$

where $f_n(x) := \sum_{k=0}^{n+2025} \left(x + \sqrt{x^2 - 1} \cos\left(\frac{2\pi k}{n + 2026} \right) \right)^{n+26}$.

Problem 4

$$\int_0^\infty \int_0^\infty \sin\left(x^2 + y^2\right) \, \mathrm{d}x \, \mathrm{d}y$$

Problem 5

$$\int_1^\infty \frac{1}{\lfloor x + \lfloor x \rfloor \rfloor^2} \, \mathrm{d}x$$

Problem 6

$$\int \frac{f^{-1}(x)}{1 + f(x)} \, \mathrm{d}x \,,$$

where $f(x) := x^\varphi$, $\varphi = \dfrac{1+\sqrt{5}}{2}$ is the golden ratio.

Problem 7

$$\int_0^{\frac{\pi}{2}} \ln\left(\left|\sin(2025!x)\right|\right) \ln\left(\left|\sin(2026!x)\right|\right) \mathrm{d}x$$

Problem 8

$$\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \sin(x)^{\tan(x)-2} \left(\tan^2(x) \ln(\sin(x)) + \sin^2(x)\right) \mathrm{d}x$$

Problem 9

$$\int_e^{2e^2} \frac{e^{W(x)}}{x^2}\,\mathrm{d}x\,,$$

where $W(x)$ is the Lambert W function, i.e. the inverse function of $f(x) = xe^x$.

Problem 10

$$\int_0^{2\pi} \frac{\sin^{67}(x)}{2026 - \cos(x)}\,\mathrm{d}x$$

Problem 11

$$\int_{-1}^{0} \left(\frac{3x^2+2x}{x^3+x^2-1} + \frac{6x^2+2x}{2x^3+x^2-2}\right)\sqrt{(x^3+x^2-1)(2x^3+x^2-2)}\,\mathrm{d}x$$

Problem 12

$$\int_3^7 \frac{10 - \{x\}}{\lfloor x \rfloor + \lfloor 10 - x \rfloor + 10}\,\mathrm{d}x$$

Problem 13

$$\int_1^\infty x\sqrt{x^2-1}e^{-x^2}\,\mathrm{d}x$$

Problem 14

$$\int_1^\infty \sum_{k=1}^\infty \frac{1}{k+1}\left\lfloor \frac{k+1}{\{x\}}\right\rfloor^{-\lfloor x\rfloor}\mathrm{d}x$$

Problem 15

$$\lim_{n\to\infty}\int_0^1 \sin\left(\frac{\pi}{2n}\lfloor nx+1\rfloor\right)\mathrm{d}x$$

Problem 16

$$\left\lfloor \int_2^{2025} \frac{1}{\lfloor 2\log_x(5)+4\log_x(3)\rfloor}\,\mathrm{d}x\right\rfloor$$

Problem 17

$$\int x^{x\mathrm{d}x},$$

where $\int f(x)^{\mathrm{d}x}$ is the product integral.

Note: The definite product integral $\int_a^b f(x)^{\mathrm{d}x}$ is defined analogously to the Riemann integral with product in place of sum; that is, by the limit

$$\lim_{\Delta x_i \to 0} \prod_{i=1}^{n} f(x_i)^{\Delta x_i} \qquad \text{where } a = x_0 < x_1 < \cdots < x_n = b \text{ and } \Delta x_i = x_i - x_{i-1}.$$

The indefinite product integral is then also defined analogously to the indefinite Riemann integral.

Problem 18

$$\int_0^\infty \frac{\sin(x)}{x(\cosh(x) + \cos(x))}\,\mathrm{d}x$$

Problem 19

$$\lim_{n\to\infty} n \int_0^1 \frac{1 + x^3}{n + x + x^3 + x^5}\,\mathrm{d}x$$

Problem 20

$$\int_{-1}^{1} \frac{1+\mathrm{Li}_2(x)}{4+\mathrm{Li}_2(x^2)}\,\mathrm{d}x\,,$$

where $\mathrm{Li}_2(x) := -\int_0^x \frac{\ln(1-t)}{t}\,\mathrm{d}t$ is the dilogarithm.

Problem 21

$$\int_{-\infty}^{\infty} e^{x-\sinh^2(x)}\,\mathrm{d}x$$

Problem 22

$$\int_{\sqrt{2}-1}^{\sqrt{2}+1} \frac{1}{x\left(e+(x^2+1)^{\frac{1}{\ln(x)}}\right)}\,\mathrm{d}x$$

Problem 23

$$\int_0^1 \frac{\left(\ln\left(1-x^n\right)\right)^m}{x}\,\mathrm{d}x \qquad \text{for positive integers } m, n.$$

Problem 24

$$\int_0^\infty \frac{x^{\frac{e}{2}-1}(1+\ln(x))}{x^e+1}\,\mathrm{d}x$$

Problem 25

$$\int_0^1 \frac{1-x}{1+x}\cdot\frac{1}{\sqrt{x+x^2+x^3}}\,\mathrm{d}x$$

Problem 26

$$\int_0^{\frac{\pi}{2}} \frac{1}{1+\sin^2(x)\cos^2(x)}\,\mathrm{d}x$$

Problem 27

$$\int_{-1}^{1} e^{f^{-1}(x)}\,\mathrm{d}x\,,$$

where $f(x)=\operatorname{erf}(x):=\dfrac{2}{\sqrt{\pi}}\displaystyle\int_0^x e^{-t^2}\,\mathrm{d}t$ is the error function integral.

Problem 28

$$\int_0^\infty \frac{x^2-1}{x^2+1} \cdot \frac{\sin(x)}{x}\,\mathrm{d}x$$

Problem 29

$$\prod_x (2025x + 2026),$$

where $\prod_x$ is the discrete multiplicative integral, i.e. an inverse operator of $g(f(x)) = \dfrac{f(x+1)}{f(x)}$.

Problem 30

$$\int_0^\infty \frac{\sin(x)\ln(x)}{x}\,\mathrm{d}x$$